VIMON14-var4

Aetos Montelukast Tablet 10mg

DESCRIPTION
Square, beige film- coated tablet, shallow convex, bevel-edged and "HD" embossed on one face.

COMPOSITION
Montelukast Sodium which is equivalent to Montelukast 10 mg.

PHARMACODYNAMICS
The cysteinyl leukotrienes (LTC_4, LTD_4, LTE_4) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils. These important pro-asthmatic mediators bind to cysteinyl leukotriene (CysLT) receptors. The CysLT type-1 ($CysLT_1$) receptor is found in the human airway (including airway smooth muscle cells and airway macrophages) and on other pro-inflammatory cells (including eosinophils and certain myeloid stem cells). CysLTs have been correlated with the pathophysiology of asthma and allergic rhinitis. In asthma, leukotriene-mediated effects include bronchoconstriction, mucous secretion, vascular permeability, and eosinophil recruitment. In allergic rhinitis, CysLTs are released from the nasal mucosa after allergen exposure during both early- and late-phase reactions and are associated with symptoms of allergic rhinitis. Intranasal challenge with CysLTs has been shown to increase nasal airway resistance and symptoms of nasal obstruction.

Montelukast is a potent, orally active compound that significantly improves parameters of asthmatic inflammation. Based on biochemical and pharmacological bioassays, it binds with high affinity and selectivity to the $CysLT_1$ receptor (in preference to other pharmacologically important airway receptors such as the prostanoid, cholinergic, or ß-adrenergic receptor). Montelukast potently inhibits physiologic actions of LTC_4, LTD_4 and LTE_4 at the $CysLT_1$ receptor without any agonist activity.

PHARMACOKINETICS
Absorption: Montelukast is rapidly absorbed following oral administration. For the 10 mg film-coated tablet, the mean peak plasma concentration (C_{max}) is achieved 3 hours (T_{max}) after administration in adults in the fasted state. The mean oral bioavailability is 64%. The oral bioavailability and Cmax are not influenced by a standard meal. Safety and efficacy were demonstrated in clinical trials where the 10 mg film-coated tablet was administered without regard to the timing of food ingestion.

For the 5 mg chewable tablet, the C_{max} is achieved 2 hours after administration in adults in the fasted state. The mean oral bioavailability is 73%. Food does not have a clinically important influence with chronic administration.

Distribution: Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of Montelukast averages 8-11 litres. Studies in rats with radiolabelled Montelukast indicate minimal distribution across the blood-brain barrier. In addition, concentrations of radiolabelled material at 24 hours post-dose were minimal in all other tissues.

Metabolism: Montelukast is extensively metabolised. In studies with therapeutic doses, plasma concentrations of metabolites of Montelukast are undetectable at steady state in adults and children. In vitro studies using human liver microsomes indicate that cytochrome P450 3A4, 2A6 and 2C9 are involved in the metabolism of Montelukast. Based on further in vitro results in human liver microsomes, therapeutic plasma concentrations of Montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6. The contribution of metabolites to the therapeutic effect of Montelukast is minimal.

Plasma Protein Binding: Plasma protein binding is more than 99% (Very high).

Elimination: The plasma clearance of Montelukast averages 45 ml/min in healthy adults. Following an oral dose of radiolabelled Montelukast, 86% of the radioactivity was recovered in 5-day faecal collections and <0.2% was recovered in urine. Coupled with estimates of Montelukast oral bioavailability, this indicates that Montelukast and its metabolites are excreted almost exclusively via the bile. It is not known whether Montelukast is removable by peritoneal dialysis or hemodialysis. The pharmacokinetics of Montelukast is nearly linear at doses of up to 50 mg.

INDICATIONS
- For the prophylaxis and chronic treatment of asthma in adults and pediatric patients 12 months of age and older.
- Montelukast is indicated in adults and pediatric patients 2 years of age and older for the relief of daytime and nighttime symptoms of seasonal allergic rhinitis.

CONTRAINDICATIONS
Hypersensitivity to the active substance or to any of the excipients.

WARNINGS AND PRECAUTIONS
- The efficacy of oral Montelukast for the treatment of acute asthma attacks has not been established. Therefore, oral Montelukast should not be used to treat asthma attacks.
- Patients should be advised never to use oral Montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled ß-agonist should be used. Patients should seek their doctors' advice as soon as possible if they need more inhalations of short-acting ß-agonists than usual.
- Montelukast should not be substituted for inhaled or oral corticosteroids.
- There are no data demonstrating that oral corticosteroids can be reduced when Montelukast is given concomitantly. In rare cases, patients on therapy with anti-asthma agents including Montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These cases usually, but not always, have been associated with the reduction or withdrawal of oral corticosteroid therapy. The possibility that leukotriene receptor antagonists may be associated with emergence of Churg-Strauss syndrome can neither be excluded nor established. Physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients.
- Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.
- Treatment with Montelukast does not alter the need for patients with aspirin-sensitive asthma to avoid taking aspirin and other non-steroidal anti-inflammatory drugs.
- Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose- galactose malabsorption should not take this medicine.

Effects on ability to drive and use machines
Montelukast is not expected to affect a patient's ability to drive a car or operate machinery. However, in very rare cases, individuals have reported drowsiness or dizziness.

PREGNANCY AND LACTATION
Pregnancy
The safety and efficacy of montelukast have not been determined in pregnant women. Rarely, congenital limb defects have been reported during post-marketing surveillance; however, causality has not been established. Due to the lack of human safety information, montelukast should be used in pregnant women only if the potential benefit outweighs the potential risk to the fetus.

Lactation
Lactation studies with montelukast in humans have not been conducted. Montelukast is excreted into the milk of lactating rats. Until further data are available, caution is advised if montelukast is used in nursing women. Available evidence and/or expert consensus is inconclusive or is inadequate for determining infant risk when used during breastfeeding. Weigh the potential benefits of drug treatment against potential risks before prescribing this drug during breastfeeding.

DRUG INTERACTIONS
- Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma. In drug-interactions studies, the recommended clinical dose of montelukast did not have clinically important effects on the pharmacokinetics of theophylline, prednisone, prednisolone, oral contraceptives (ethinyl estradiol/ norethindrone 35/1), terfenadine, digoxin and warfarin.
- Montelukast is metabolised by CYP 3A4, caution should be exercised, particularly in children, when montelukast is co-administered with inducers of CYP 3A4, such as phenytoin, phenobarbital and rifampicin.
- *In vitro* studies have shown that montelukast is a potent inhibitor of CYP 2C8. However, data from a clinical drug-drug interaction study involving montelukast and rosiglitazone (a probe substrate representative of drugs primarily metabolized by CYP 2C8) demonstrated that montelukast does not inhibit CYP 2C8 in vivo. Therefore, montelukast is not anticipated to markedly alter the metabolism of drugs metabolised by this enzyme (e.g., paclitaxel, rosiglitazone, and repaglinide.)

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MAIN SIDE/ADVERSE EFFECTS

The following side/adverse effects have been selected on the basis of their potential clinical significance (possible signs and symptoms in parentheses where appropriate) - not necessarily inclusive:

Those indicating for medical attention:

- ***Incidence less frequent:***
  Elevated hepatic enzymes (asymptomatic) which include alanine aminotransferase and aspartate transferase.
- ***Incidence rare:***
  Pyuria (pus in the urine)

Those indicating need for medical attention only if they continue or are bothersome:

- ***Incidence more frequent:***
  Headache
- ***Incidence less frequent:***
  Abdominal or stomach pain, asthenia or fatigue (weakness or unusual tiredness), cough, dental pain, dizziness, dyspepsia (heartburn), fever, gastroenteritis, infectious (abdominal or stomach pain), nasal congestion (stuffy nose) and skin rash.

The following adverse reactions have been reported in post-marketing use of montelukast:

- ***Blood and lymphatic system disorders:***
  Thrombocytopenia.
- ***Psychiatric disorders:***
  Obsessive-compulsive symptoms, dream abnormalities including nightmares, hallucinations, psychomotor hyperactivity (including irritability, restlessness, agitation including aggressive behaviour and tremor), depression, suicidal thinking and behaviour (suicidality) in very rare cases, insomnia.
- ***Immune system disorders:***
  Hypersensitivity reactions including anaphylaxis, hepatic eosinophilic infiltration.
- ***Nervous system disorders:***
  Dizziness, drowsiness, paraesthesia/hypoesthesia, seizure.
- ***Cardiac disorders:***
  Palpitations.
- ***Gastro-intestinal disorders:***
  Diarrhoea, dry mouth, dyspepsia, nausea, vomiting.
- ***Hepatobiliary disorders:***
  Elevated levels of serum transaminases (ALT, AST), cholestatic hepatitis.
- ***Skin and subcutaneous tissue disorders:***
  Angioedema, bruising, urticaria, pruritus, rash, erythema nodosum
- ***Musculoskeletal and connective tissue disorders:***
  Arthralgia, myalgia including muscle cramps.
- ***General disorders and administration site conditions:***
  Asthenia/fatigue, malaise, oedema.

Very rare cases of Churg-Strauss Syndrome (CSS) have been reported during montelukast treatment in asthmatic patients.

OVERDOSE

Symptoms

Symptoms included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.

Treatment

Treatment is symptomatic and supportive. Treatment may include removal of unabsorbed material from the gastrointestinal tract, clinical monitoring and supportive therapy if required. It is not known if montelukast can be removed by peritoneal dialysis or hemodialysis.

DOSAGE AND ADMINISTRATION

Oral.

Montelukast should be taken once daily. For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs. Patients with both asthma and allergic rhinitis should take only one dose daily in the evening.

Adults 15 Years of Age and Older with Asthma and/or Seasonal Allergic Rhinitis

The dosage for adults 15 years of age and older is one 10 mg tablet daily.

Pediatric Patients 6 to 14 Years of Age with Asthma and/or Seasonal Allergic Rhinitis

The dosage for pediatric patients 6 to 14 years of age is one 5 mg chewable tablet daily.

Pediatric Patients 2 to 5 Years of Age with Asthma and/or Seasonal Allergic Rhinitis

The dosage for pediatric patients 2 to 5 years of age is one 4 mg chewable tablet daily or one packet of 4 mg oral granules daily.

Pediatric Patients 12 months to 2 Years of Age with Asthma

The dosage for pediatric patients 12 months to 2 years of age is one packet of 4 mg oral granules daily.

General Recommendations

The therapeutic effect of Montelukast on parameters of asthma control occurs within one day. Montelukast tablets, chewable tablets and oral granules can be taken with or without food. Patients should be advised to continue taking Montelukast while their asthma is controlled, as well as during periods of worsening asthma.

No dose adjustment is necessary for pediatric patients, for the elderly, for patients with renal insufficiency, or mild-to-moderate hepatic impairment, or for patients of either gender.

Montelukast is a long term-controller medication which should not be substituted for short acting beta-agonists. It is effective alone or in combination with other prophylactic agent.

Montelukast is a preventive agent, which should be used in addition to other drugs for the management of asthma.

Therapy with Montelukast in Relation to Other Treatments for Asthma

Montelukast can be added to a patient's existing treatment regimen.

Reduction in Concomitant Therapy:

Bronchodilator Treatments: Montelukast can be added to the treatment regimen of patients who are not adequately controlled on bronchodilator alone. When a clinical response is evident (usually after the first dose), the patient's bronchodilator therapy can be reduced as tolerated.

Inhaled Corticosteroids: Treatment with Montelukast provides additional clinical benefit to patients treated with inhaled corticosteroids. A reduction in the corticosteroid dose can be made as tolerated. The dose should be reduced gradually with medical supervision. In some patients, the dose of inhaled corticosteroids can be tapered off completely. Montelukast should not be abruptly substituted for inhaled corticosteroids.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage:
Store below 30°C. Protect from light and moisture.

Presentation/Packing:
Alu-alu blister of 3 x10's, 10 x 10's.

Product Registration Holder: Hovid Pharmacy Sdn. Bhd.
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Manufactured by: HOVID Bhd.
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31200 Chemor, Malaysia.

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